

# Problem 9

## Problem 9

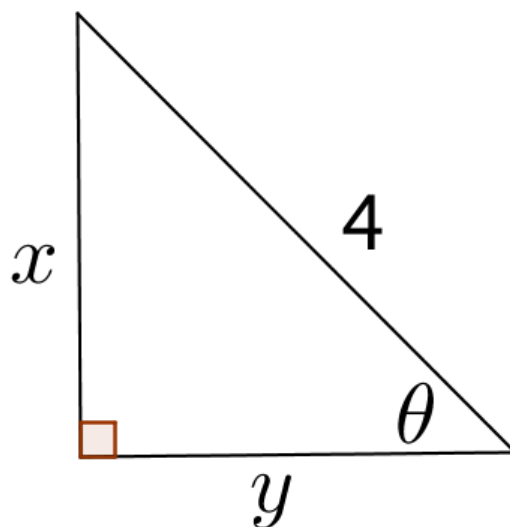
**Problem 1** Simplify each of the following expressions.

(a)  $\cos^{-1}(\sin(\pi/2))$

**Explanation.** By the unit circle,  $\sin(\pi/2) = 1$ , and so we are looking for  $\cos^{-1}(1)$ . The range of  $\cos^{-1}$  is  $[0, \pi]$ , and so, by properties of inverse functions,  $\cos^{-1}(1) = 0$ .

(b)  $\tan(\sin^{-1}(x/4))$

**Explanation.** Let  $\theta = \sin^{-1}(x/4)$ , then  $\sin(\theta) = x/4$ . We can then draw the corresponding right triangle:



Calling the adjacent side  $y$ , by the Pythagorean Theorem we obtain

$$4^2 = x^2 + y^2 \implies y = \sqrt{16 - x^2}$$

Remark: Since  $\theta$  is in the range of  $\sin^{-1}$ , it follows that  $-\pi/2 \leq \theta \leq \pi/2$ . On the other hand, the expression  $\tan(\sin^{-1}(x/4))$  is defined only for

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$-4 < x < 4$  (Note:  $\sin^{-1}(4/4) = \frac{\pi}{2}$  and  $\tan$  is not defined at  $\frac{\pi}{2}$ ; similarly for  $x = -4$ .)

Therefore,  $-\pi/2 < \theta < \pi/2$ , which implies that  $\cos(\theta) = y/4 > 0$ . Therefore,  $y > 0$ .

Then

$$\begin{aligned}\tan(\sin^{-1}(x/4)) &= \tan(\theta), \\ &= \frac{x}{y}, \\ &= \frac{x}{\sqrt{16 - x^2}}.\end{aligned}$$

Note:  $\tan(\theta)$  has the same sign as  $x$ , since  $y > 0$ .